

Exercise 14: Non-convex Optimization

Lecturer: Aurelien Lucchi

Problem 1 (Power method - example):

1. Perform 5 iterations of the unnormalized power method (i.e. $\mathbf{x}_k = \mathbf{A}\mathbf{x}_{k-1}$) to approximate the dominant eigenvector of the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & -12 \\ 1 & -5 \end{pmatrix}$$

To do so, take $\mathbf{x}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ as the initialization point. What is the dominant eigenvector that is being approximated? (hint: the entries of the vector are integers).

2. What is the eigenvalue corresponding to the dominant eigenvector?
3. What is the other eigenvalue of $\mathbf{A}$?

Problem 2 (Asymptotic convergence power iteration):

Prove the following asymptotic convergence result for the power method:

$$\mathbf{w}_k = \frac{\hat{\mathbf{A}}^k \mathbf{w}_0}{\|\hat{\mathbf{A}}^k \mathbf{w}_0\|} = \left(\frac{\lambda_1}{|\lambda_1|} \right)^k \frac{c_1}{|c_1|} \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} + r_k$$

where $\|r_k\| \rightarrow 0$ as $k \rightarrow \infty$.

Hint: Use the decomposition of the starting vector $\mathbf{w}_0$ discussed in class: $\mathbf{w}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$.

Problem 3 (Power method - angular convergence):

Given a matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_d$. Consider the power method update $\mathbf{w}^{(k+1)} = \frac{\mathbf{A}\mathbf{w}^{(k)}}{\|\mathbf{A}\mathbf{w}^{(k)}\|}$. We define the angle between two vectors as $\angle(\mathbf{w}, \mathbf{u}) := \arccos \frac{\langle \mathbf{w}, \mathbf{u} \rangle}{\|\mathbf{u}\| \|\mathbf{w}\|}$.

We can write $\mathbf{w}^{(k)}$ in the eigenbasis $\{\mathbf{w}_i\}_{i=1}^d$: $\mathbf{w}^{(k)} = \sum_{i=1}^d \alpha_i^{(k)} \mathbf{w}_i$. Crucially, we note that

$$\tan(\angle(\mathbf{w}^{(k)}, \mathbf{w}_1)) = \frac{\sqrt{\sum_{j=2}^d (\alpha_j^{(k)})^2}}{\alpha_1^{(k)}}. \quad (1)$$

1. Show that $\tan(\angle(\mathbf{w}^{(k+1)}, \mathbf{w}_1)) \leq \frac{\lambda_2}{\lambda_1} \tan(\angle(\mathbf{w}^{(k)}, \mathbf{w}_1))$.

Problem 4 (Convergence for PL functions):

Under the same finite-sum setting discussed in Problem 1, assume that f is μ -PL with L -Lipschitz continuous gradients and bounded gradients ($\|\nabla f_i(\mathbf{x}_k)\|^2 \leq B^2$).

1. Prove that

$$\mathbb{E}[f(\mathbf{x}_{k+1}) - f^*] \leq (1 - 2\eta_k \mu)[f(\mathbf{x}_k) - f^*] + \frac{LB^2 \eta_k^2}{2}.$$

2. Let $\delta_f(k) \equiv k^2 \mathbb{E}[f(\mathbf{x}_k) - f^*]$. Using $\eta_k = \frac{2k+1}{2\mu(k+1)^2}$, show that

$$\delta_f(k+1) \leq \delta_f(k) + \frac{LB^2}{2\mu^2},$$

3. Conclude that

$$\mathbb{E}[f(\mathbf{x}_{k+1}) - f^*] \leq \frac{LB^2}{2\mu^2(k+1)}.$$